MODULE 1 · LECTURE 23

Big Data and the Internet of Things (IoT) in Agriculture: Use Cases and Benefits

What makes data "big", how an IoT system senses, communicates, processes and acts, which wireless technologies suit which farm, how sensor data are cleaned and turned into decisions — with worked examples on data volume, battery life and an irrigation trigger, and a program that cleans a soil-moisture data stream.

Learning objectives

After this lecture you should be able to:

1. Define big data and explain its characteristics (the Vs) with farm examples.
2. Describe the Internet of Things and the layers of an IoT system.
3. Compare wireless communication technologies used in farm IoT systems.
4. Explain edge and cloud computing, data cleaning and the main kinds of big-data analytics.
5. Estimate data volume, sensor battery life and a sensor-based irrigation trigger.
6. Describe use cases, benefits and challenges of big data and IoT in agriculture, including Indian conditions.

Scope: earlier lectures introduced the Internet (Lecture 7), databases (Lecture 6), sensors and actuators (Lecture 12) and soil-water budgeting (Lecture 13). This lecture joins them into IoT systems that collect large amounts of farm data. How AI learns from such data is taken up in Lectures 24–28, and India's national digital-agriculture platforms in Lectures 30–31.

1. What is big data?

Farms, machines, satellites and markets now produce data continuously. De Mauro et al. (2016), after reviewing many definitions, proposed that **big data** is the information asset characterised by such high volume, velocity and variety that it requires specific technology and analytical methods to be transformed into value. Size alone does not make data "big": its speed, its many forms and the effort needed to analyse it matter as much (Gandomi & Haider, 2015).

Big data: data so large, fast and varied that special technology and analytics are needed to turn it into value

Volume	Velocity	Variety	Veracity	Value
how much data	*how fast it arrives*	*how many kinds*	*how trustworthy*	*what it is worth*
a drone survey of 20 ha ≈ 2.7 GB per flight	40 soil sensors send 3,840 messages a day	numbers, images, text, GPS tracks, prices	spikes, stuck probes, missing messages	decisions: when to irrigate, spray, sell

Volume, velocity and variety are the original three Vs; veracity and value are commonly added. Farm examples from this lecture.

Figure 1. The Vs of big data, with farm examples. Original diagram.

Characteristic	Meaning	Farm example
Volume	The amount of data	Thousands of drone and satellite images of a district each season
Velocity	The rate at which data are produced and must be handled	Soil, weather and machine sensors reporting every few minutes
Variety	The many forms and sources of data	Sensor numbers, images, videos, farmer's text and voice queries, GPS tracks, market prices
Veracity	Accuracy, dependability and trustworthiness of the data	A faulty probe, a missing message, a wrong price entry
Value	The usefulness of the insights hidden in the data	Knowing when to irrigate, when disease risk is high, when to sell

The original three Vs were later extended with veracity and value (Siddique et al., 2025). Much big data is **unstructured** — text, images, audio and video rather than neat tables — and Gandomi and Haider (2015) estimate that such data make up 95% of big data. They also warn that noisy, heterogeneous data invite pitfalls such as **spurious correlation**: patterns that appear by chance.

Big data in smart farming

Wolfert et al. (2017) reviewed big data in smart farming and found that it reaches beyond the farm to the whole food supply chain. Big data are used to give **predictive insights** into farm operations, to drive **real-time operational decisions** and to redesign business processes. The same review expects big data to shift roles and power among the players in the food chain, and it gives research priority to **data governance** — how farm data are controlled and shared — while protecting privacy.

> **Worked example 1 — how much data does a 20 ha farm produce?**
> Sensors: 40 soil-moisture nodes, each sending a 24-byte message every 15 minutes (96 a day).
> Messages per day = 40 × 96 = 3,840 — one every 22.5 seconds on average (velocity).
> Bytes per day = 3,840 × 24 = 92,160 ≈ 92 kB; per year ≈ 92,160 × 365 ≈ 33.6 MB.
> Drone: a 20-megapixel camera (5,472 × 3,648 pixels) at 2 cm per pixel covers 109.4 m × 73.0 m ≈ 0.80 ha per image. With 75% forward and 70% side overlap, each new image adds only 0.80 × 0.25 × 0.30 ≈ 0.060 ha.
> Images for 20 ha ≈ 20 ÷ 0.060 ≈ 334; at about 8 MB each ≈ 2.7 GB for one flight (volume).
> One drone survey therefore holds as much data as about 80 years of the soil-sensor messages: images drive volume, sensors drive velocity.

The sensor, camera and file-size values in worked examples 1 and 2 are illustrative assumptions chosen to show the method; real values depend on the equipment used.

2. The Internet of Things (IoT)

The **Internet of Things** is a network of physical objects — sensors, machines, animals' tags, valves — that can sense, communicate over the Internet and act with little human involvement. In agriculture, IoT brings together earlier technologies such as wireless sensor networks, radio-frequency identification (RFID), cloud computing, middleware and end-user applications (Elijah et al., 2018), so that farms can be monitored and maintained with minimal human involvement (Farooq et al., 2019).

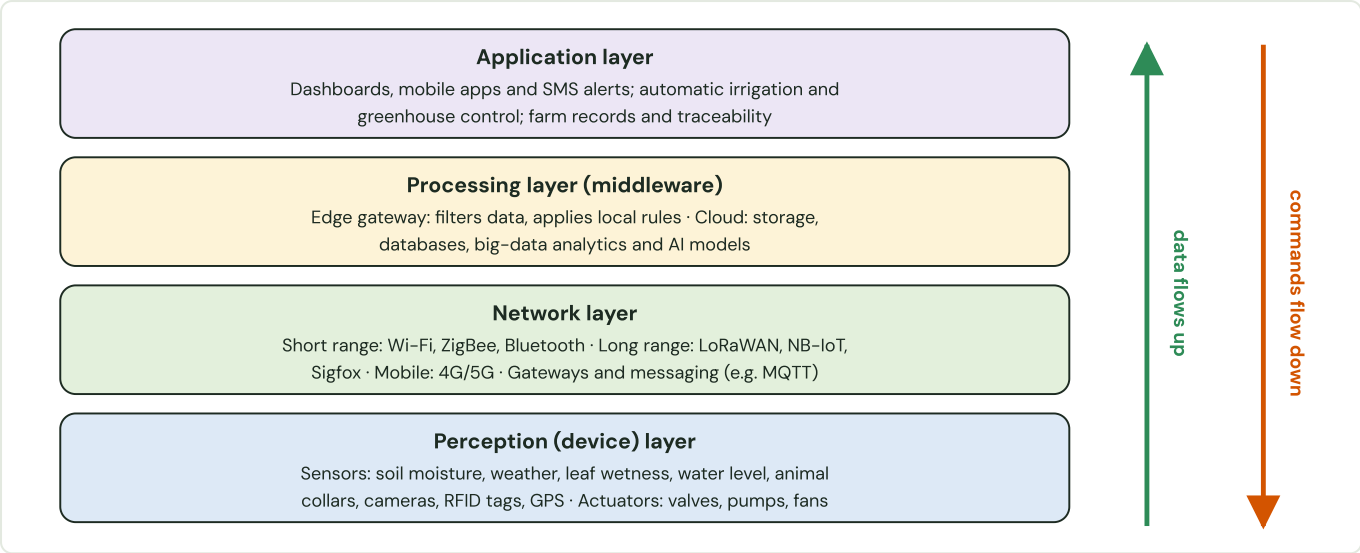


Figure 2. Layers of a farm IoT system: data flow up from devices to applications; commands flow back down to actuators. Original diagram.

Layer	Role	Examples
Perception (device)	Senses the field and acts on it	Soil-moisture, temperature, humidity, rain, leaf-wetness and water-level sensors; cameras; RFID ear tags; GPS; valves, pumps and fans (Lecture 12)
Network	Carries data between devices, gateways and the Internet	Wi-Fi, ZigBee, Bluetooth, LoRaWAN, NB-IoT, mobile networks; gateways; lightweight messaging such as MQTT
Processing (middleware)	Stores, cleans and analyses data; applies rules and models	Microcontroller boards and gateways at the edge; cloud storage, databases (Lecture 6) and analytics
Application	Presents results and controls the farm	Dashboards, mobile apps (Lecture 14), SMS alerts, automatic irrigation and greenhouse control

Low-cost hardware has made such systems affordable. In a review of IoT irrigation systems, García et al. (2020) found that Arduino boards were the most used nodes — the Arduino UNO alone in 34 papers — and noted that Arduino-type nodes offer a low-cost option for developing countries and smaller farms. Raspberry Pi boards were used where more processing power was needed.

3. Communication technologies

Choosing the radio link is a trade-off between **range**, **data rate** and **power use**. Short-range standards usually reach up to about 100 m (García et al., 2020); low-power wide-area networks (**LPWAN**) such as LoRaWAN, Sigfox and NB-IoT send small amounts of data over long distances. Mekki et al. (2019) found that Sigfox and LoRa are better for battery life, capacity and cost, while NB-IoT is better for latency and quality of service.

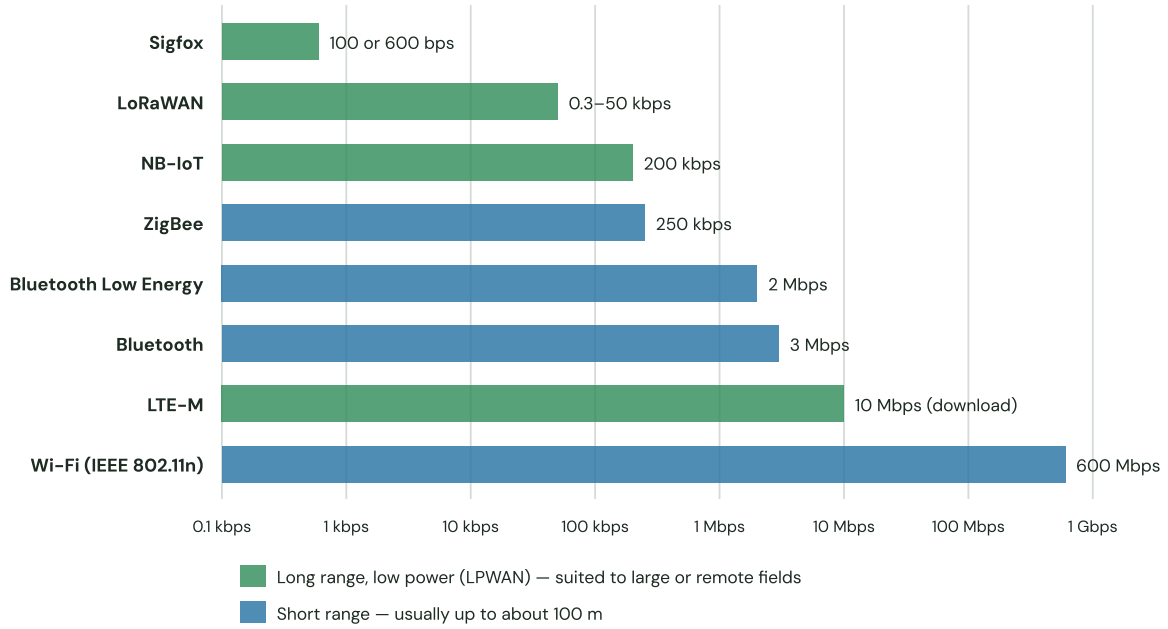


Figure 3. Maximum data rates of wireless technologies used in farm IoT (logarithmic scale). Data rates from García et al. (2020); original chart.

Technology	Range	Data rate	Typical farm use
Wi-Fi	Short (up to about 100 m)	Up to 600 Mbps (802.11n)	Greenhouses, farmhouse, cameras, gateways with mains power
ZigBee (IEEE 802.15.4 mesh)	Short; extended by mesh hops	Up to 250 kbps	Networks of sensor nodes in greenhouses and orchards
Bluetooth / Bluetooth Low Energy	Short	Up to 3 Mbps / 2 Mbps	Reading handheld sensors and animal tags with a smartphone
LoRaWAN	Long	0.3–50 kbps	Soil and weather sensors spread over large or remote fields; in India it uses the 865–867 MHz band
Sigfox	Long	100 or 600 bps	Very small, infrequent messages such as tank levels or alarms
NB-IoT / LTE-M	Long (mobile-network coverage)	200 kbps / up to 10 Mbps	Devices where mobile networks reach: meters, trackers, storage monitors
GSM / 4G / 5G	Long (mobile-network coverage)	High	Gateways, cameras, SMS alerts and pump controllers

Data rates and the Indian LoRaWAN band are from García et al. (2020). In the irrigation systems they reviewed, Wi-Fi was the most used technology (63 papers), followed by GSM (25) and ZigBee (23); LoRa was noted as a good option for secluded areas with no service.

Worked example 2 — how long will a sensor battery last?
A node wakes every 15 minutes (900 s), reads its sensor for 2 s at 10 mA and sends one message. Battery: 2,400 mAh usable.
LoRa node: transmits for 1 s at 40 mA; sleeps 897 s at 0.010 mA.
Charge per cycle = 0.010 × 897 + 10 × 2 + 40 × 1 = 8.97 + 20 + 40 = 68.97 mA·s → average current = 68.97 ÷ 900 = 0.0766 mA.
Life = 2,400 ÷ 0.0766 ≈ 31,300 h ≈ **3.6 years**.
Wi-Fi node: joins the network and sends for 4 s at 120 mA; sleeps 894 s at 0.020 mA.
Charge per cycle = 0.020 × 894 + 20 + 120 × 4 = 517.9 mA·s → average 0.575 mA → life ≈ 4,170 h ≈ **174 days (about 6 months)**.
The radio dominates the energy budget: a short, low-power transmission lets a field node run for years without a battery change.

4. From data to decisions

Edge and cloud computing

Data can be processed at the **edge** — on the node or a gateway near the field — or in the **cloud**, in remote data centres. Edge processing reacts quickly, sends less data and keeps working when the Internet link fails; the cloud offers large storage, heavy analytics and access from anywhere. Most farm systems combine the two: the gateway applies simple rules (open the valve), and the cloud keeps the history and runs models.

Cleaning the data (veracity)

Raw sensor data always contain errors: missing messages when the radio link drops, impossible values (spikes) from a splash or loose wire, and **stuck** readings when a probe fails. Data accuracy, dependability and trustworthiness are what veracity means (Siddique et al., 2025). Simple rules catch most problems: reject missing values, values outside the physically possible range, and long runs of identical readings. Rules must be tuned so that good data are not thrown away.

Analysing the data

Kind of analytics	Question answered	Farm example
Descriptive	What happened?	Daily rainfall, soil moisture and yield maps
Predictive	What will happen?	Disease risk from leaf wetness and temperature; yield forecast (Lecture 26)
Prescriptive	What should be done?	When and how much to irrigate; variable-rate fertiliser maps (Lecture 12)

Large data sets are stored in databases and distributed storage (Lecture 6) and analysed with statistics and machine learning (Lectures 19 and 24–28). Big-data analytics needs methods that are efficient on massive, noisy, mixed data (Gandomi & Haider, 2015).

5. Worked example: a soil-moisture IoT system

Worked example 3 — setting the irrigation trigger (FAO-56 soil-water budget, Lecture 13)
Loam: field capacity θFC = 0.30 and wilting point θWP = 0.14 m³/m³; root zone Zr = 0.6 m; depletion fraction p = 0.5, a value commonly used for many crops (Allen et al., 1998).
TAW = 1000 × (0.30 − 0.14) × 0.6 = 96 mm; RAW = 0.5 × 96 = 48 mm.
Trigger moisture = θFC − RAW ÷ (1000 × Zr) = 0.30 − 48 ÷ 600 = **0.22 m³/m³**.
When a sensor reports 0.22, the net irrigation needed is 48 mm; with drip at 90% efficiency the gross depth is 48 ÷ 0.9 ≈ 53 mm, or about 533 m³ of water per hectare.

The program below simulates ten days of hourly readings from one soil-moisture sensor in each of three zones, adds three typical faults — a radio dropout, a spike and a stuck probe — then cleans the data and decides when each zone needs water.

Cleaning a soil-moisture data stream and triggering irrigation in Python (Lecture 8)

```
import random

FC, WP, ROOT_MM, P = 0.30, 0.14, 600, 0.5   # loam, 0.6 m root zone, FAO-56 depletion fraction
TRIGGER = FC - P * (FC - WP)                 # irrigate when soil moisture falls to 0.22 m3/m3
ET = {"A": 6.5, "B": 5.0, "C": 3.5}          # crop water use of each zone, mm/day

# 1. Simulate 10 days of hourly readings from one soil-moisture sensor per zone
random.seed(23)
stream, theta = [], {z: FC for z in ET}
for hour in range(240):
    for z in ET:
        theta[z] -= ET[z] / 24 / ROOT_MM     # soil dries as the crop uses water
        value = round(theta[z] + random.gauss(0, 0.002), 3)
        if z == "B" and hour == 100:
            value = 0.97                     # spike: water splash on the probe
        if z == "A" and 60 <= hour < 72:
            value = None                     # radio dropout: no message received
        if z == "C" and 150 <= hour < 162:
            value = 0.262                    # probe stuck at one value
        stream.append((hour, z, value))

# 2. Clean the data (veracity): reject missing, impossible and stuck readings
clean = {z: [] for z in ET}
rejected = {"missing": 0, "out of range": 0, "stuck": 0}
last, repeats = {}, {z: 0 for z in ET}
for hour, z, v in stream:
    if v is None:
        rejected["missing"] += 1
        continue
    if not 0.05 <= v <= 0.50:
        rejected["out of range"] += 1
        continue
    repeats[z] = repeats[z] + 1 if v == last.get(z) else 0
    last[z] = v
    if repeats[z] >= 5:                      # 6th identical value in a row
        rejected["stuck"] += 1
        continue
    clean[z].append((hour, v))

# 3. Decide: irrigate when the average of the last 3 valid readings reaches the trigger
print(f"Readings received: {len(stream)}  rejected: {rejected}")
for z in ET:
    for i in range(2, len(clean[z])):
        hour = clean[z][i][0]
        avg = sum(v for _, v in clean[z][i - 2:i + 1]) / 3
        if avg <= TRIGGER:
            net = (FC - avg) * ROOT_MM
            print(f"Zone {z}: trigger on day {hour // 24 + 1} at {hour % 24:02d}:00 "
                  f"(moisture {avg:.3f}) -> irrigate {net:.0f} mm net, {net / 0.9:.0f} mm gross (drip)")
            break
    else:
        print(f"Zone {z}: lowest moisture {min(v for _, v in clean[z]):.3f} -> no irrigation needed
yet")

Output:
Readings received: 720  rejected: {'missing': 12, 'out of range': 1, 'stuck': 7}
Zone A: trigger on day 8 at 12:00 (moisture 0.219) -> irrigate 48 mm net, 54 mm gross (drip)
Zone B: trigger on day 10 at 10:00 (moisture 0.220) -> irrigate 48 mm net, 54 mm gross (drip)
Zone C: lowest moisture 0.241 -> no irrigation needed yet
```

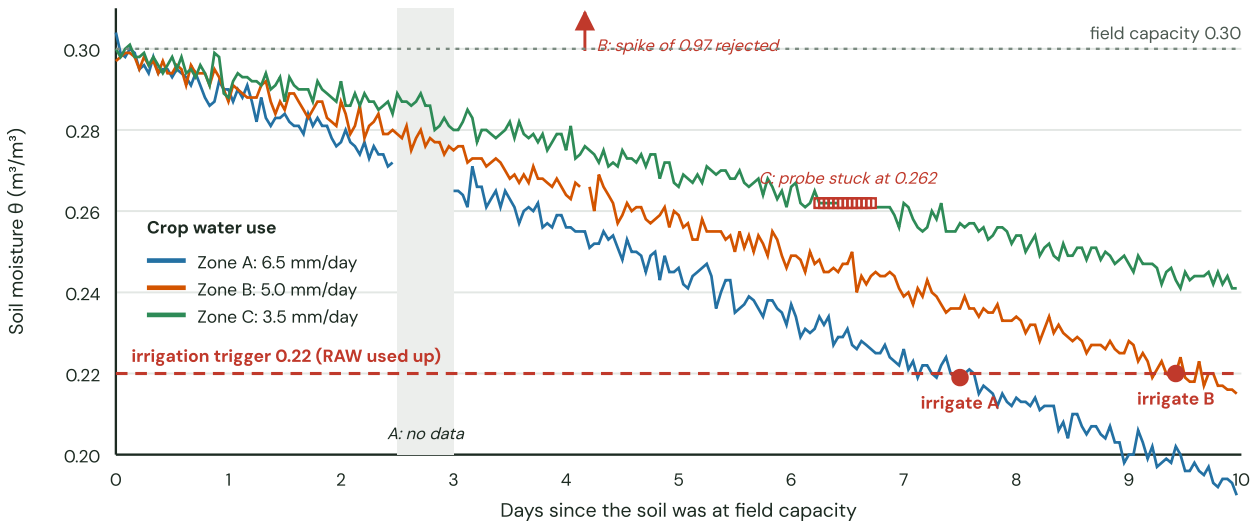


Figure 4. Cleaned soil-moisture readings from the program, with the faults it found and the irrigation triggers. Original chart generated from the program's data.

- **Missing**: 12 readings from zone A (a 12-hour radio dropout) were skipped.
- **Out of range**: the 0.97 spike in zone B was rejected — soil cannot hold that much water.
- **Stuck**: zone C's probe repeated 0.262 for 12 hours; the last 7 readings were rejected. The first 5 were accepted, because a stuck probe can only be recognised after several identical values.
- **Decisions**: zone A, whose crop uses the most water, reached the trigger at 12:00 on day 8 (7.5 days after field capacity); zone B at 10:00 on day 10; zone C did not need water within the 10 days.

The program was run and the output shown is its actual output. In an earlier run with a stricter rule — rejecting a fourth identical value instead of a sixth — the program also rejected two genuine readings, which repeated by chance because moisture changes slowly and readings are rounded. Data-cleaning rules must be tested on real data before use.

6. Use cases

Use case	Data collected	Decision or benefit
Smart irrigation	Soil moisture, weather, flow and water level	Irrigate only when and where needed, automatically or by alert (García et al., 2020)
Weather stations and disease forecasting	Temperature, humidity, rainfall, leaf wetness	Warnings when weather favours disease or pests; timely, targeted spraying (Lecture 25)
Greenhouse climate control	Air temperature, humidity, light, CO_2	Fans, foggers, vents and shade screens switched automatically (Lecture 12)
Livestock monitoring	Wearable sensors on collars and tags; sound and image analysis	Real-time health monitoring and early disease detection; better welfare and lower production costs (Neethirajan, 2017, 2020)
Storage and cold chain	Temperature and humidity in godowns, cold stores and refrigerated trucks	Alarms before produce spoils; records for quality and traceability
Machinery and yield monitoring	GPS position, fuel, engine data, yield-sensor readings	Yield maps, field-efficiency analysis, maintenance alerts (Lecture 26)
Remote sensing archives	Satellite and drone images over many seasons	Crop condition, area and damage assessment over large regions (Lecture 15)
Supply chain and markets	Prices, arrivals, transport and storage records	Price forecasts, traceability and new business models (Wolfert et al., 2017; Lecture 14)

7. Benefits and challenges

Benefits	Challenges
Water, fertiliser and pesticide applied when and where needed	Patchy mobile and Internet coverage in rural fields
Early warnings: frost, disease-favourable weather, sick animals	Power supply in the field: batteries, solar panels, radio choice
Less labour: fields and animals checked remotely from a phone	Cost, calibration and maintenance of sensors, especially for small farms
Predictive insights and real-time decisions (Wolfert et al., 2017)	Data quality (veracity): missing, noisy and faulty readings
Records for traceability, certification, credit and insurance	Interoperability: devices and platforms that do not share standards
Better animal welfare and lower production costs (Neethirajan, 2020)	Data governance, sharing and privacy (Wolfert et al., 2017)
Low-cost boards and sensors put basic systems within reach of small farms (García et al., 2020)	Cyber-security of devices and networks (Farooq et al., 2019); digital skills

In India, most farms are small and fragmented, so an IoT system often makes more sense when it is shared — by a farmer producer organisation, a custom-hiring centre or a water-users' group — or offered as a service. National programmes that build shared digital infrastructure for agriculture are discussed in Lectures 30 and 31.

8. Key takeaways

- Big data is data with such volume, velocity and variety that special technology and analytics are needed to extract value; veracity and value are commonly added Vs.
- In farming, images drive data volume while sensors drive velocity: one 20 ha drone survey can equal decades of soil-sensor messages.
- An IoT system has perception, network, processing and application layers; data flow up and commands flow down.
- Short-range radios (Wi-Fi, ZigBee, Bluetooth) carry more data; LPWAN (LoRaWAN, Sigfox, NB-IoT) trades data rate for range and battery life.
- Edge computing acts fast and locally; the cloud stores history and runs heavy analytics.
- Sensor data must be cleaned of missing, impossible and stuck readings before decisions are made.
- Use cases include irrigation, disease warning, greenhouses, livestock, storage, machinery, remote sensing and markets.
- Connectivity, power, cost, data quality, standards, data governance and security are the main challenges.

9. Examination questions

Short-answer questions (2 marks)

Q1. Define big data.

Information so high in volume, velocity and variety that it needs specific technology and analytical methods to be turned into value (De Mauro et al., 2016).

Q2. Name the five Vs of big data.

Volume, velocity, variety, veracity and value.

Q3. What is the Internet of Things?

A network of physical objects (sensors, machines, tags, actuators) that sense, communicate over the Internet and act with little human involvement.

Q4. Name the four layers of an IoT system.

Perception (device), network, processing (middleware) and application.

Q5. What is LPWAN? Give two examples.

Low-power wide-area network: long-range, low-data-rate radio for battery devices — e.g. LoRaWAN, Sigfox, NB-IoT.

Q6. State the maximum data rates of ZigBee and LoRaWAN.

ZigBee up to 250 kbps; LoRaWAN 0.3–50 kbps.

Q7. Distinguish edge and cloud computing.

Edge: processing on or near the device, fast and local; cloud: processing and storage in remote data centres.

Q8. What is meant by data veracity?

The accuracy, dependability and trustworthiness of data.

Q9. Give two faults found in sensor data streams.

Any two of: missing messages, out-of-range spikes, stuck readings, drift.

Q10. Give the irrigation trigger for $\theta FC = 0.30$, $\theta WP = 0.14$, $Zr = 0.6$ m, $p = 0.5$.

RAW = 48 mm; trigger $\theta = 0.30 - 48/600 = 0.22$ m³/m³.

Long-answer questions (5–10 marks) — answer outlines

Q1. Explain big data and its characteristics with agricultural examples.

Definition; five Vs with Figure 1 and the table; unstructured data; worked example 1; smart-farming uses (Wolfert et al., 2017).

Q2. Describe the architecture of an IoT system for agriculture.

Figure 2; four layers with roles and examples; data up, commands down; low-cost nodes.

Q3. Compare wireless communication technologies used in farm IoT systems.

Range, data rate and power; Figure 3 and the table; LPWAN trade-offs; worked example 2 battery life.

Q4. Explain how sensor data are turned into decisions, with an irrigation example.

Edge and cloud; cleaning (veracity); descriptive, predictive, prescriptive analytics; worked example 3; program and Figure 4.

Q5. Discuss use cases of big data and IoT in agriculture.

Irrigation, disease forecasting, greenhouses, livestock, storage, machinery, remote sensing, markets.

Q6. Discuss the benefits and challenges of IoT and big data for Indian agriculture.

Benefits and challenges table; small holdings and shared services; data governance and security.

10. Glossary

Term	Meaning
Actuator	Device that acts on the field: valve, pump, fan, motor
Big data	Data whose volume, velocity and variety need special technology and analytics
Cloud computing	Storage and processing in remote data centres over the Internet
Data governance	Rules on who controls, shares and uses data
Edge computing	Processing data on or near the device that produces it
Gateway	Device linking field nodes to the Internet
IoT (Internet of Things)	Network of objects that sense, communicate and act
LoRaWAN	Long-range, low-power wide-area network protocol
LPWAN	Low-power wide-area network
MQTT	Lightweight publish–subscribe messaging protocol used in IoT
NB-IoT	Narrowband IoT over mobile networks
RFID	Radio-frequency identification, e.g. animal ear tags
Spurious correlation	A pattern in data that arises by chance
Unstructured data	Text, images, audio and video not organised in tables
Veracity	Accuracy and trustworthiness of data
Wireless sensor network (WSN)	Network of battery-powered sensor nodes linked by radio

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All diagrams, examples and programs are original. Equipment values in the worked examples are illustrative; every calculation and the program output were checked by running them.

Practice MCQs

Q1. The three original Vs of big data are

(a) volume, velocity and variety
(b) value, veracity and volume
(c) velocity, validity and visibility
(d) volume, variability and value

Q2. The veracity of data refers to its

(a) size
(b) speed of arrival
(c) accuracy and trustworthiness
(d) number of formats

Q3. In the layered IoT architecture, soil-moisture sensors and irrigation valves belong to the

(a) application layer
(b) perception (device) layer
(c) network layer
(d) business layer

Q4. Which technology gives long range at a low data rate and low power?

(a) Wi-Fi
(b) Bluetooth
(c) LoRaWAN
(d) Ethernet

Q5. The maximum data rate of ZigBee is about

(a) 250 kbps
(b) 50 Mbps
(c) 600 Mbps
(d) 100 bps

Q6. In the review of IoT irrigation systems by García et al. (2020), the most used communication technology was

(a) LoRa
(b) Wi-Fi
(c) Sigfox
(d) Z-Wave

Q7. In India, LoRaWAN operates in the

(a) 865–867 MHz band
(b) 5.8 GHz band
(c) 2.4 GHz band only
(d) 60 GHz band

Q8. With θFC = 0.30, θWP = 0.14, a 0.6 m root zone and p = 0.5, the readily available water is

(a) 96 mm
(b) 48 mm
(c) 24 mm
(d) 160 mm

Q9. A soil-moisture probe that reports exactly the same value for many hours most likely

(a) is stuck or faulty
(b) shows heavy rain
(c) shows perfect irrigation
(d) has a faster radio

Q10. Processing data on a gateway near the field rather than in a distant data centre is called

(a) cloud computing
(b) edge computing
(c) batch processing
(d) data warehousing

Q11. Wolfert et al. (2017) identified which issue as a research priority for big data in smart farming?

(a) Governance of data and data sharing
(b) Shortage of sensors
(c) Too few satellites
(d) Slow tractors

Q12. In the worked example, one 20 ha drone survey (about 2.7 GB) equals roughly how many years of data from 40 soil sensors?

(a) 1 year
(b) 8 years
(c) 80 years
(d) 800 years

Answers

Q1 — (a) volume, velocity and variety
Q2 — (c) accuracy and trustworthiness
Q3 — (b) perception (device) layer
Q4 — (c) LoRaWAN
Q5 — (a) 250 kbps
Q6 — (b) Wi-Fi
Q7 — (a) 865–867 MHz band
Q8 — (b) 48 mm
Q9 — (a) is stuck or faulty
Q10 — (b) edge computing
Q11 — (a) Governance of data and data sharing
Q12 — (c) 80 years